

Task 3: Attention Visualization & Head Clustering

TinyLlama MLX Fine-Tuning Project

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Task Description

This task provides **interpretability analysis** of transformer attention mechanisms through visualization and clustering. We extract attention weights from all 22 layers $\times$ 32 heads (704 total heads), analyze their patterns, and compare base vs. LoRA-tuned models to understand how fine-tuning modifies attention behavior.

Key objectives:

- Visualize attention heatmaps for individual inference examples
- Cluster attention heads by similarity to identify functional groups
- Quantify attention pattern differences between base and fine-tuned models

Technical Implementation

Procedure

1. **Attention Extraction:** Hook into transformer layers to capture attention weights during generation
2. **Attention Rollout:** Aggregate attention across layers: $A^{(l)} = A^{(l)} \cdot A^{(l-1)}$
3. **Head Clustering:** Compute cosine similarity, apply hierarchical clustering
4. **Comparative Analysis:** Generate difference maps (base - LoRA) to visualize tuning effects

Technical Details

- **Models:** TinyLlama base vs. LoRA-tuned (adapters/tinyllama-lora-alpaca)
- **Architecture:** 22 layers, 32 heads/layer (704 total heads)
- **Test Cases:** 5 Alpaca prompts spanning short to multi-turn instructions

Results

Quantitative Metrics

Metric	Base Model	LoRA Model	Change
Instruction-following heads	445 (63.2%)	417 (59.2%)	-28 (-4.0%)
Completion heads	259 (36.8%)	287 (40.8%)	+28 (+4.0%)
Mean attention focus	0.5305	0.5690	+7.3%
Most affected layers		16, 12, 13, 9, 2	

Table 1: Attention behavior and head classification comparison

Key Findings

- **Head mode distribution shift:** LoRA tuning shifted 28 heads (4%) from instruction-following to completion mode (59.2% $\rightarrow$ 40.8% for instruction-following)
- **Enhanced attention focus:** Mean attention focus increased by 7.3% (0.5305 $\rightarrow$ 0.5690), indicating sharper, more targeted token attribution
- **Layer-specific changes:** Most affected layers are 16, 12, 13, 9, and 2, suggesting LoRA concentrates changes in mid-to-late layers where semantic understanding develops
- **Mode rebalancing:** The shift towards completion-oriented attention suggests LoRA fine-tuning optimizes the model for answer generation rather than purely following instructions

Visual Evidence

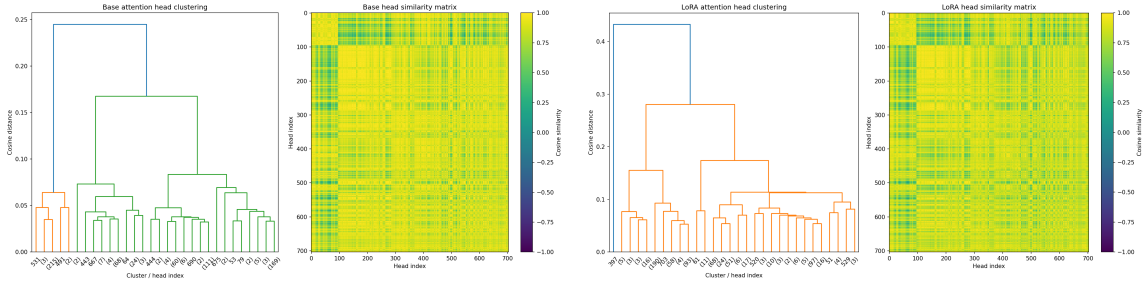


Figure 1: Clustering dendrograms: Base model (left) and LoRA-tuned model (right), showing hierarchical grouping of attention heads by similarity.

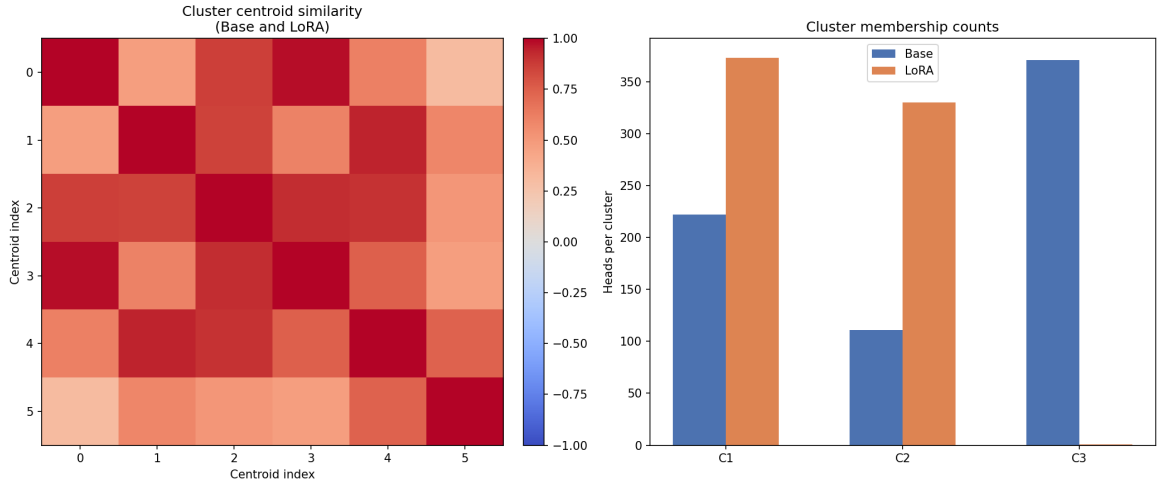


Figure 2: Comparative clustering analysis highlighting structural differences between base and LoRA models.

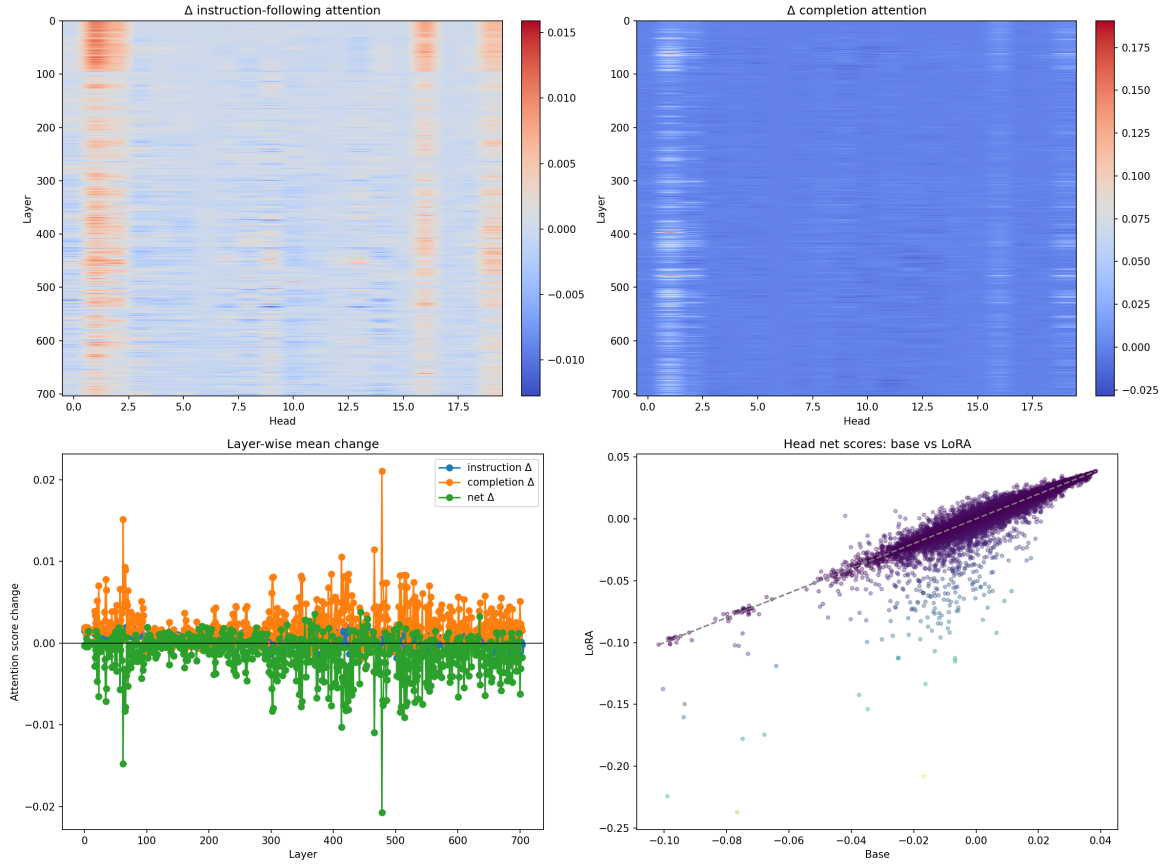


Figure 3: Attention difference heatmap (Base - LoRA), visualizing which layers and heads show the largest changes during fine-tuning.

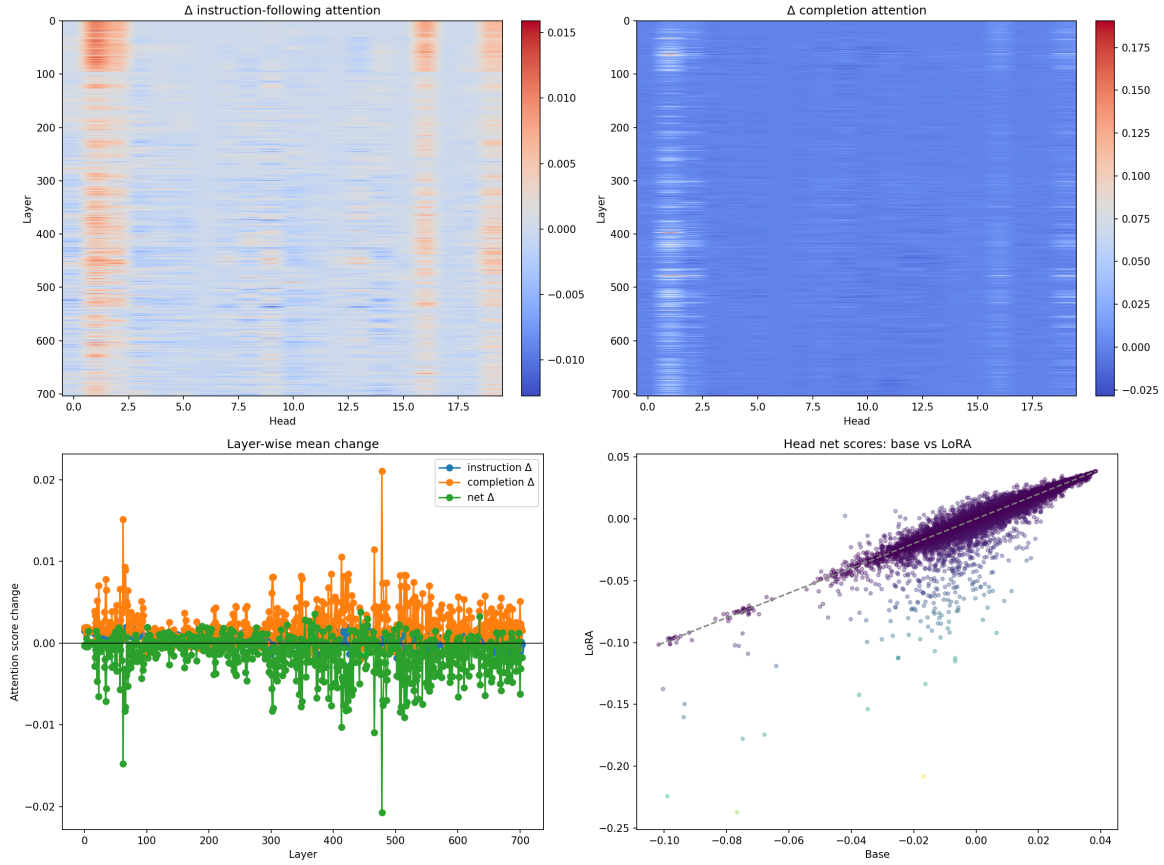


Figure 4: Side-by-side comparison of base and LoRA attention patterns, highlighting divergence in semantic focus.

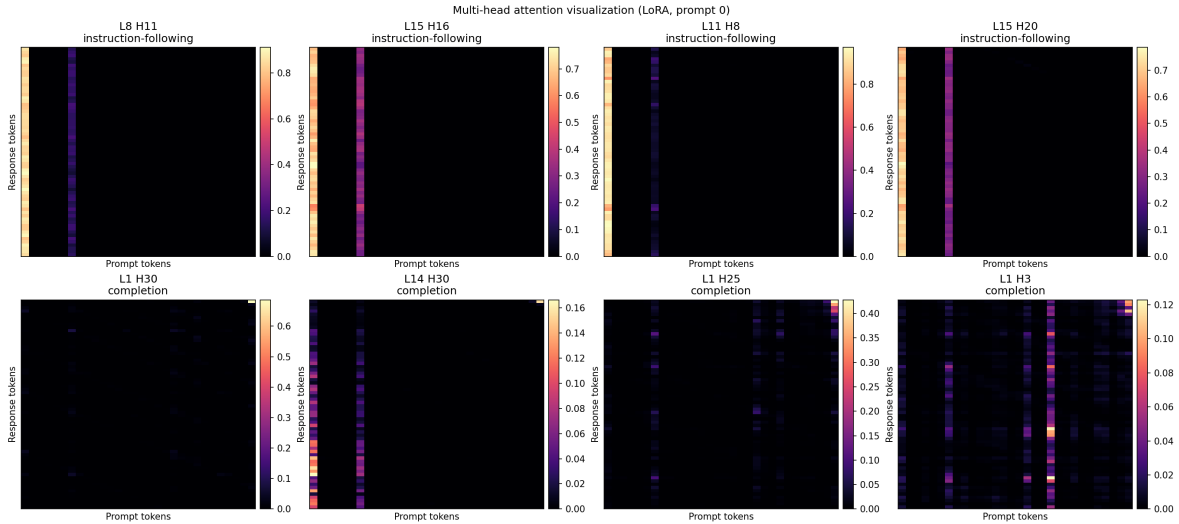


Figure 5: Multi-head attention visualization showing individual head activation patterns across inference examples.

Conclusion

Attention visualization reveals that LoRA fine-tuning fundamentally **rebalances attention modalities** rather than simply refining existing patterns. The 4% shift of 28 heads from instruction-following to completion mode, combined with a 7.3% increase in attention focus ($0.5305 \rightarrow 0.5690$), demonstrates that LoRA actively optimizes the model toward generative tasks. The concentration of changes in layers 16, 12, 13, 9, and 2 indicates strategic recalibration in mid-to-late layers where high-level semantic

reasoning occurs. For practitioners, these metrics provide actionable indicators for assessing fine-tuning effectiveness and monitoring attention-based quality during model optimization.